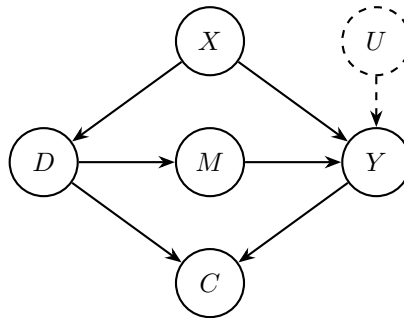


Problem Set: Directed Acyclic Graphs

BUSS975: Causal Inference in Financial Research

Problem 1 (Reading a DAG)

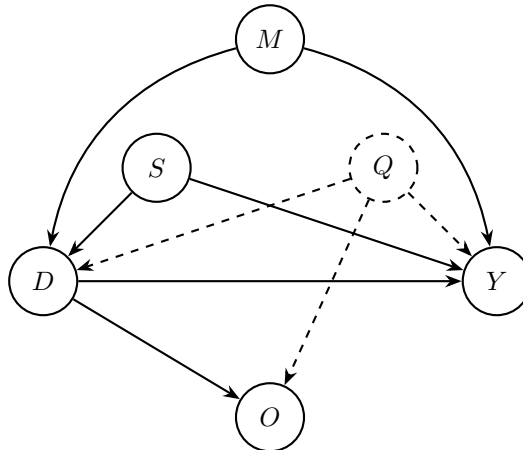
Consider the following graph, where U is unobserved.



- (a) List **every** path from D to Y and say whether each is directed (causal) or not. For each non-directed path, identify the structure at its middle node — chain, fork, or collider — and state whether the path is open or closed *before* any conditioning.
- (b) Which of these paths are **backdoor** paths? Justify using the definition (both conditions).
- (c) Is C a collider on every path it lies on? Explain why “collider” is a property of a path rather than of a variable.
- (d) Suppose a researcher conditions on $\{X, C\}$. State precisely what each does: which paths open, which close, and whether the causal effect is identified.

Problem 2 (A DAG: index inclusion)

You want the effect of **inclusion in a major equity index** (D) on **firm investment** (Y). Let S = firm size, Q = firm quality (**unobserved**), O = institutional ownership, M = market conditions.



- Enumerate every path from D to Y and classify each as directed or backdoor.
- The path $D \rightarrow O \leftarrow Q \rightarrow Y$ connects D and Y . Explain why it is **not** a backdoor path, and why it is nonetheless closed.
- Which set of variables would satisfy the backdoor criterion if everything were observable? Given that Q is *not* observable, is the effect identified by conditioning on observables? State your conclusion and what it implies for the research design.
- A referee insists you control for institutional ownership, O , “since index inclusion changes ownership and ownership affects investment.” Give **two separate** reasons this is wrong.

Problem 3 (Bad controls and selection)

- (a) Give the operational rule for identifying a bad control from the DAG, stated in terms of the backdoor criterion. Then explain why a randomly assigned treatment does **not** protect you from bad-control bias.
- (b) For each of the following, identify what the sample-construction step conditions on, and whether that variable is plausibly a collider:
 - i. A study of mutual fund performance using funds alive at the end of the sample period.
 - ii. A study of the effect of leverage on firm value using CRSP–Compustat merged data.
 - iii. An event study of merger announcements using only **completed** deals.
 - iv. A study of bond pricing using only bonds with a credit rating.
- (c) For each of the following studies, name one plausible bad control and explain the mechanism:
 - i. Effect of a disclosure mandate on cost of capital.
 - ii. Effect of a bank merger on small-business lending.
 - iii. Effect of a governance reform on firm value.
- (d) A paper reports its main specification with and without a questionable control, and the coefficient moves substantially. What should the authors conclude, and what should they *not* conclude?